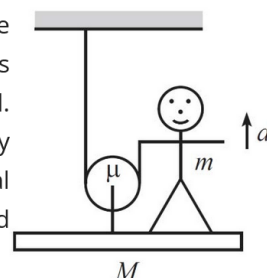


You have 60 minutes to respond to the questions and submit your answers.

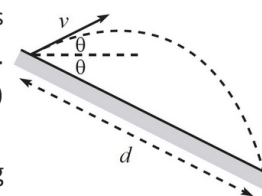
1. Consider a bullet shot underwater with the initial velocity $\mathbf{v}_0$ experiencing a resistive force $\mathbf{F} = -km\mathbf{v}$ as it moves. Show that the speed of the bullet at any moment is given by, $v = v_0 - kx$ with x representing the displacement since the start of the motion. **(score : 15)**

2. A person stands on a platform and pulley system, as shown in the figure. The platform, person, and pulley masses are M , m , and μ , respectively. The rope is massless. Let the person pull up on the rope so that she accelerates a upward. (Assume that the platform is somehow constrained to stay level, perhaps by having the ends run along some rails.) Find the tension in the rope, the normal force between the person and the platform, and the tension in the rod connecting the pulley to the platform (ignore rotational dynamics). **(score : 35)**



3. **(a)** For a given initial speed, at what inclination angle should a ball be thrown so that it travels the maximum horizontal distance by the time it returns to the ground? Assume that the ground is horizontal, and that the ball is released from ground level. **(score : 15)**
(b) What is the optimal angle if the ground is sloped upward at an angle β (or downward, if β is negative)? **(score : 15)**

4. A plane tilts down at an angle θ below the horizontal. On this plane, a projectile is fired with speed v at an angle θ above the horizontal, as shown in the figure. What is the distance, d , along the plane that the projectile travels? **(score : 15)**
 What is d in the limit $\theta \rightarrow 90^\circ$? **(score : 5)**



5. **(Extra Credit - score : 20)** Consider a particle of mass m and charge q entering a space with magnetic field of strength B in the Y direction of the Cartesian Coordinate system as shown in the figure. Recall that the motion of a particle in magnetic field is described by the force $\mathbf{F}_B = q\mathbf{v} \times \mathbf{B} = m\mathbf{a}$ in which $\mathbf{v}$ represents the velocity of the particle at any given moment.

Show that the location of this particle as a function of time t can be written as,

$$x - x_0 = C \cos(Dt)$$

$$y - y_0 = \dot{y}_0 t$$

$$z - z_0 = C \sin(Dt)$$

$$\mathbf{B} = B_0 \hat{j}$$

$$C = \frac{\frac{dz_0}{dt} m}{qB_0}$$

$$D = \frac{qB_0}{m}$$

